

- 6 The photograph shows a ground squirrel.



(Source: Caroline Wilcox)

magnification $\times 0.5$

Ground squirrels feed on a variety of foods including nuts.

Ground squirrels have evolved food pouches.

The photograph shows the food pouches of a ground squirrel.

food pouch



(Source: © All Canada Photos/Alamy Stock Photo)

The table shows some information about three types of nut: acorns, hazelnuts and walnuts.

Type of nut	Photograph	Description	Energy content / kJ per 100 g
acorn		1.0 to 3.0 cm long 0.8 to 2.0 cm wide	1619
hazelnut		1.5 to 2.5 cm long 1.0 to 1.5 cm wide hard covering	2788
walnut		3.0 to 5.0 cm long 2.0 to 4.0 cm wide hard shell	2822

(Source: © Science photo library, Source: © Jon Stokes/Science photo library, Source: © Science photo library)

(a) The volume of the smallest walnut is approximately 50 cm³.

Calculate the approximate volume of the largest hazelnut, using the formula

$$\text{volume} = \frac{4}{3} \pi l w^2$$

where $\pi = 3$, l = length of the hazelnut and w = width of the hazelnut.

(1)

Answer cm³

(b) A student investigated the feeding preferences of ground squirrels.

- (i) Design a laboratory investigation to find out which of these three types of nut ground squirrels prefer to eat.

(3)

- (ii) Predict the results of this investigation, giving reasons.

Use the information in the table to support your answer.

(3)

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings present.

(c) Suggest how ground squirrels evolved pouches.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 6 = 10 marks)
